Oral Exam 1 Information
REAL VARIABLES I
Fall 2026

Last updated: September 27, 2026.

0.1 Scheduling and exam format

You must schedule an exam appointment during the week before Fall Break or during the days after the break.
When you arrive for the exam, you will be given a choice of two topics from the list below. You will select one topic and have up to 30 minutes to prepare your exposition. This preparation period is closed book: you may not use notes, books, or other resources.
After the preparation period, you will have an approximately 30-minute discussion at the board. You may consult any notes you produced during your preparation time.
During the board discussion, you should be prepared to:

- give a coherent overview of the selected topic and explain how its main ideas fit together;
- state definitions and mathematical statements precisely;
- present the central proofs at the level of their overall structure;
- supply detailed arguments for individual steps when asked; and
- respond to questions about hypotheses, examples, counterexamples, and connections among the results.

A presentation does not have to be flawless to earn full marks. The goal is to demonstrate that you have understood the topic and can think about it at the board. Tips or suggestions may be provided if you get stuck. If you are receptive to suggestions and recover from a temporary hesitation, you can still earn full marks.

0.2 Example grading rubric

The exam will be assessed holistically. The following descriptions illustrate the level of performance associated with each score range.

- **90–100%:** Demonstrates a strong and connected understanding of the topic. Definitions and statements are precise when requested, the main proof ideas are correct, and detailed arguments are supplied with only minor omissions or errors. Responds thoughtfully to questions and can use a tip to recover from hesitation. The presentation need not be perfect.
- **80–89%:** Demonstrates good understanding of the main ideas and proof structure. Most definitions and statements are accurate, but there are some gaps, imprecisions, or errors of substance. Responds productively to questions and suggestions, though some issues remain unresolved.
- **70–79%:** Demonstrates substantial but incomplete understanding. Can explain several central ideas and parts of the proofs, but important definitions, hypotheses, connections, or proof steps are missing or incorrect. Follow-up questions reveal uneven command of the topic. Can carry out at least one technical proof to completion.

- **60–69%:** Demonstrates limited understanding. Can state some relevant definitions or results and make partial progress on proofs, but the overall structure is unclear or contains errors. Substantial prompting is needed to proceed.
- **50–59%:** Demonstrates limited understanding. Some relevant facts are recalled, but central definitions, statements, and proof ideas are seriously incomplete or inaccurate. Makes little progress even with significant guidance.
- **Below 50%:** Does not yet demonstrate a workable understanding of the selected topic.

0.3 Topics

Status: The contents of the topics in this list is finalized. Additional topics may be added as the course progresses.
Unless an item is explicitly marked *statement only*, you should be able to give precise definitions and prove both the statement and auxiliary facts for the result. This applies to all mentioned facts: theorems, propositions, lemmata, or corollaries, and examples.

0.3.1 Topic 1: Outer measures, measures, and covering constructions

1. **Definitions and distinctions.**
   - σ-algebra, measurable space, measure, finite measure, and outer measure;
   - finite additivity, countable additivity, finite subadditivity, and countable subadditivity;
   - increasing and decreasing sequences of sets;
   - the outer measure ℓ^* generated by a covering family $\mathcal{A}$ and a cost function $\ell\colon \mathcal{A} \to [0, +\infty]$;
   - Lebesgue outer measure $\mathcal{L}_2^*$ generated by the area cost on open axis-parallel rectangles in $\mathbb{R}^2$.

   Be able to distinguish carefully between a measure, whose domain is a σ-algebra, and an outer measure, whose domain is the full power set.
2. **Outer measure generated by covering costs.** The generated outer measure ℓ^* is indeed an outer measure.
3. **Extenision property** When does an outer measure generated by a covering cost extend the cost.
4. **Finite additivity and σ-subadditivity** Explain how they give σ additivity
5. **Basic properties and continuity of measures.** For a measure μ, prove finite additivity, monotonicity, the difference formula, countable subadditivity, countable superadditivity, and continuity from below. Prove continuity from above under the hypothesis $\mu(E_1) < +\infty$.
6. **Countable additivity from continuity below.** Prove that a finitely additive set function is countably additive if it is continuous from below. Identify precisely where finite additivity and continuity from below are used.
7. **Lebesgue outer measure agrees with area on rectangles.** State

   $$\mathcal{L}_2^*(R) = \operatorname{area}(R)$$

   for every open axis-parallel rectangle $R \subseteq \mathbb{R}^2$. Prove the statement, including the reduction to a compact inner rectangle, the finite-subcover reduction, and the finite-grid area comparison.

8. **Counterexamples.** Give and explain a counterexample to continuity from above when the finiteness hypothesis $\mu(E_1) < +\infty$ is omitted. Be able to explain why an outer measure need not be finitely additive on its full domain.

Course notes: Outer Measures; σ-algebras, Measures, and Their Properties.

0.3.2 Topic 2: Measurable functions and sigma-algebra constructions

1. **Definitions and constructions.** Define:
   - measurable spaces and measurable maps;
   - inverse images and the pullback σ-algebra;
   - the σ-algebra generated by a collection of sets;
   - the Borel σ-algebra $\mathcal{B}(Y)$ of a topological space Y;
   - the product σ-algebra $\mathcal{B} \otimes \mathcal{C}$;
   - indicator functions and non-negative measurable simple functions.
2. **Generator criterion for measurability.** If $\mathcal{B} = \sigma(\mathcal{G})$, prove that $f\colon (X, \mathcal{A}) \to (Y, \mathcal{B})$ is measurable if and only if $f^{-1}(G) \in \mathcal{A}$ for every $G \in \mathcal{G}$.
3. **Continuous maps are Borel measurable.** Prove that a continuous map between topological spaces is measurable with respect to their Borel σ-algebras.
4. **Measurability of a paired map.** For measurable spaces $(Y, \mathcal{B})$ and $(Z, \mathcal{C})$, prove that $(f, g)\colon X \to Y \times Z$ is $(\mathcal{A}, \mathcal{B} \otimes \mathcal{C})$-measurable if and only if both coordinate functions are measurable.
5. **Increasing simple-function approximation.** For measurable $f\colon X \to [0, +\infty]$, define
$$s_n(x) = 2^{-n} \lfloor 2^n \min\{f(x), n\} \rfloor .$$
Prove that every s_n is a non-negative measurable simple function and that $s_n \uparrow f$ pointwise. If f is bounded, also prove that $s_n \to f$ uniformly.
6. **Generators of Euclidean Borel sets.** Explain why rational open cubes, or equivalently rational open rectangles, generate $\mathcal{B}(\mathbb{R}^d)$. Identify where countability is used.
7. **Operations and pointwise limits of measurable functions.** State the closure results for algebraic operations, finite and countable lattice operations, and pointwise limits of real-valued measurable functions. Prove these statements by reducing to measurable level sets or to composition with Borel maps.
8. **The Vitali set and the obstruction to measuring every subset of $\mathbb{R}$.** Define the equivalence relation
$$x \sim y \quad \Longleftrightarrow \quad x - y \in \mathbb{Q},$$
and construct a set $V \subseteq [0, 1]$ containing one representative from each equivalence class that meets $[0, 1]$. Prove that the rational translates of V are pairwise disjoint and that a countable collection of these translates covers $[0, 1]$ while remaining inside a bounded interval. Use translation invariance and countable additivity to prove that V cannot be Lebesgue measurable. Explain how this obstruction motivates restricting a measure to a σ-algebra and why it is related to the failure of finite additivity of Lebesgue outer measure on all subsets of $\mathbb{R}$.

Course notes: σ-algebras, Measures, and Their Properties; Measurable Functions; The Non-negative Lebesgue Integral.

0.3.3 Topic 3: The Carathéodory construction and uniqueness of measures

1. **Definitions and the construction chain.**

   - Carathéodory measurability and the Carathéodory splitting criterion;
   - the Carathéodory σ-algebra $\mathcal{M}_{\mu^*}$ associated with an outer measure μ^*;
   - complete measures and completion;
   - π-systems and Dynkin (λ-)systems.

   Describe the construction chain

   $$\text{rectangle areas} \longrightarrow \mathcal{L}_2^* \longrightarrow \mathcal{M}_{\mathcal{L}_2^*} \longrightarrow \lambda_2.$$

2. **Carathéodory's theorem.** For an outer measure μ^*, prove that $\mathcal{M}_{\mu^*}$ is a σ-algebra and that $\mu^*|_{\mathcal{M}_{\mu^*}}$ is a countably additive measure. Prove also that this measure is complete. Your proof should include complements, finite unions, the finite splitting identity, countable unions, and countable additivity on disjoint measurable sets.

3. **Dynkin's π–λ theorem (statement only).** If $\mathcal{P}$ is a π-system, then the smallest Dynkin system containing $\mathcal{P}$ is $\sigma(\mathcal{P})$. No proof is required.

4. **Coincidence criterion for finite measures.** Let $\mathcal{P}$ be a π-system and $\mathcal{A} = \sigma(\mathcal{P})$. If finite measures μ and ν agree on $\mathcal{P}$ and satisfy $\mu(X) = \nu(X)$, prove that $\mu = \nu$ on $\mathcal{A}$. Construct the coincidence class, prove that it is a Dynkin system, and invoke Dynkin's theorem. Explain exactly where finiteness is used.

5. **Characterization of Lebesgue measure on $\mathbb{R}$.** Prove that every translation-invariant Borel measure μ on $\mathbb{R}$ that is finite on compact sets has the form $\mu = c\lambda$ for some $c \geq 0$. Your proof should first determine the measure of half-open intervals and then use uniqueness on a generating π-system. Positivity on every nonempty open interval implies $c > 0$; the normalization $\mu([0,1)) = 1$ gives $\mu = \lambda$.

6. **Borel sets are Carathéodory measurable for Lebesgue outer measure.** Prove that bounded open rectangles are Carathéodory measurable and then use the generated-σ-algebra argument to prove

   $$\mathcal{B}(\mathbb{R}^2) \subseteq \mathcal{M}_{\mathcal{L}_2^*}.$$

Course notes: Carathéodory Measurability and the Coincidence Criterion for Measures.

0.3.4 Topic 4: The non-negative Lebesgue integral

1. **Definitions and conventions.**

   - the superlevel distribution function of a non-negative measurable function;
   - the Archimedean integral of a non-increasing function;
   - the non-negative Lebesgue integral through the layer-cake formula;
   - the canonical disjoint-level-set representation of a non-negative measurable simple function;
   - the values-times-measures formula for the integral of a simple function.

   All integrals in this topic may take the value $+\infty$. Be able to explain the convention $0 \cdot (+\infty) = 0$ where it is needed.

2. **Integral of a simple function.** Explain the layer-cake formula for a non-negative measurable simple function (picture is ok, formal algebraic proof is unnecessary, but picture must be correctly labeled and explained). The values-times-measures sum is independent of the chosen indicator-function representation (argument is ok, formal algebraic proof is unnecessary).

3. **Lebesgue monotone convergence theorem.** If $0 \le f_n \uparrow f$ pointwise, prove that f is measurable and
$$\int_X f_n \, d\mu \uparrow \int_X f \, d\mu.$$
Explain the passage from increasing superlevel sets to increasing superlevel measures and then to the integral limit.

4. **Positive linearity for simple functions.** (sketch) For non-negative measurable *simple* functions φ, ψ and $a, b \ge 0$, prove
$$\int_X (a\varphi + b\psi) \, d\mu = a \int_X \varphi \, d\mu + b \int_X \psi \, d\mu.$$

5. **Positive linearity of the non-negative integral.** For non-negative measurable functions f, g and constants $a, b \ge 0$, prove
$$\int_X (af + bg) \, d\mu = a \int_X f \, d\mu + b \int_X g \, d\mu.$$
Use increasing simple-function approximations, positive linearity for simple functions, and monotone convergence. State clearly how each ingredient is used.

6. **Monotone convergence for Archimedean integrals.** State the one-dimensional monotone convergence result for non-increasing functions of the level parameter and prove it from lower sums.

Course notes: The Non-negative Lebesgue Integral.

0.3.5 Topic 5: Integral limit theorems

1. Fatou's Lemma

2. Uniform integrability of integrable functions

3. Dominated convergence theorem. Proof using Fatou's Lemma. The estaping wave and the concentrating wave counterexamples. Explanation why they do not satisfy the hypotheses of the LDC

4. Vitali's theorem: definition of tightness, equi-uniform integrability, statement and proof of Egorov's Lemma. Proof of Vitali's theorem. Discussion what hypotheses fail in the case of concentrating wave and escaping wave

5. Proof that a single integrable function is tight.